

Packet Tracer – Static and Default Route Combinations for Branch Networks

Objectives

Part 1: Build a Multi-Router Branch Network

Part 2: Configure IP Addressing

Part 3: Configure Static Routes and Default Routes

Part 4: Verify End-to-End Connectivity

Part 5: Observe Routing Behavior

Background / Scenario

You are the network administrator for a small company with a headquarters and two branch offices.

The Headquarters Router (HQ) has direct knowledge of all internal networks. The branch routers do not need to know every remote network individually, so they will use a **default route** pointing toward HQ.

This design is commonly used in real-world branch networks because it simplifies routing configuration and reduces administrative overhead.

In this lab, you will:

- Configure multiple routers
- Configure static routes
- Configure default routes
- Verify routing tables
- Test communication between remote sites
- Understand when to use static routes versus default routes

By the end of the lab, all PCs should be able to communicate across the entire network.

Instructions

Part 1: Build the Network Topology

Step 1: Add Devices

Drag the following devices into the workspace:

- 3 Routers (2911)
- 3 Switches (2960)
- 3 PCs

Rename them:

- R1-HQ
 - R2-BranchA
 - R3-BranchB
 - PC-A
 - PC-B
 - PC-C
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Step 2: Connect the Devices

LAN Connections

Device Port		Device Port	
PC-A	Fa0	S1	Fa0/1
S1	Fa0/24	R1	G0/0
PC-B	Fa0	S2	Fa0/1
S2	Fa0/24	R2	G0/0
PC-C	Fa0	S3	Fa0/1
S3	Fa0/24	R3	G0/0

Use **Copper Straight-Through** cables.

Router-to-Router Connections

Router Interface Router Interface

R1 G0/1 R2 G0/1

R1 G0/2 R3 G0/1

Use **Copper Cross-Over** cables.

Wait for all links to become green.

Part 2: Configure IP Addressing

Step 1: Configure PCs

PC-A

Setting	Value
IP Address	192.168.10.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1

PC-B

Setting	Value
IP Address	192.168.20.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.20.1

PC-C

Setting	Value
IP Address	192.168.30.10

Setting	Value
Subnet Mask	255.255.255.0
Default Gateway	192.168.30.1

Step 2: Configure Router Interfaces

R1-HQ

```
enable
configure terminal

interface g0/0
ip address 192.168.10.1 255.255.255.0
no shutdown

interface g0/1
ip address 10.1.12.1 255.255.255.252
no shutdown

interface g0/2
ip address 10.1.13.1 255.255.255.252
no shutdown
```

R2-BranchA

```
enable
configure terminal

interface g0/0
ip address 192.168.20.1 255.255.255.0
no shutdown

interface g0/1
ip address 10.1.12.2 255.255.255.252
no shutdown
```

R3-BranchB

enable

configure terminal

interface g0/0

ip address 192.168.30.1 255.255.255.0

no shutdown

interface g0/1

ip address 10.1.13.2 255.255.255.252

no shutdown

Part 3: Configure Routing

Step 1: Configure Static Routes on HQ Router

HQ knows how to reach each branch LAN.

On R1-HQ:

ip route 192.168.20.0 255.255.255.0 10.1.12.2

ip route 192.168.30.0 255.255.255.0 10.1.13.2

Meaning:

- To reach Branch A LAN, send traffic to R2.
 - To reach Branch B LAN, send traffic to R3.
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Step 2: Configure Default Route on Branch A

On R2:

ip route 0.0.0.0 0.0.0.0 10.1.12.1

Meaning:

If R2 does not know a destination network, send traffic to HQ (R1).

Step 3: Configure Default Route on Branch B

On R3:

```
ip route 0.0.0.0 0.0.0.0 10.1.13.1
```

Meaning:

If R3 does not know a destination network, send traffic to HQ (R1).

Why Use Default Routes Here?

Without default routes:

R2 would require:

```
ip route 192.168.10.0 255.255.255.0 10.1.12.1
```

```
ip route 192.168.30.0 255.255.255.0 10.1.12.1
```

R3 would require:

```
ip route 192.168.10.0 255.255.255.0 10.1.13.1
```

```
ip route 192.168.20.0 255.255.255.0 10.1.13.1
```

As networks grow, this becomes difficult to manage.

A default route replaces all those entries with one route.

Part 4: Verify Routing Tables

Step 1: Check R2

```
show ip route
```

Expected:

```
S* 0.0.0.0/0 via 10.1.12.1
```

Step 2: Check R3

```
show ip route
```

Expected:

S* 0.0.0.0/0 via 10.1.13.1

Step 3: Check R1

show ip route

Expected:

S 192.168.20.0/24 via 10.1.12.2

S 192.168.30.0/24 via 10.1.13.2

Part 5: Test Connectivity

Step 1: Ping Branch A from HQ

On PC-A:

ping 192.168.20.10

Expected:

Reply from 192.168.20.10

Step 2: Ping Branch B from HQ

ping 192.168.30.10

Expected:

Reply from 192.168.30.10

Step 3: Ping Between Branches

From PC-B:

ping 192.168.30.10

Expected:

Reply from 192.168.30.10

Traffic Path:

PC-B

↓

R2

↓

R1 (HQ)

↓

R3

↓

PC-C

Step 4: Use Traceroute

From PC-B:

```
tracert 192.168.30.10
```

Expected:

```
1 192.168.20.1
```

```
2 10.1.12.1
```

```
3 10.1.13.2
```

```
4 192.168.30.10
```

This confirms traffic is passing through HQ before reaching Branch B.

Troubleshooting Tips

✓ Verify all interfaces are up

```
show ip interface brief
```

✓ Verify routes exist

```
show ip route
```

✓ Verify PCs have correct gateways

✓ Check router-to-router IP addresses

✓ Verify cable types

✓ Ensure all interfaces have no shutdown

✓ Use:

show running-config

to verify route configuration

What You Learned

- How static routes are configured
 - How default routes work
 - Why branch networks often use default routes
 - How routers forward packets using next-hop addresses
 - How to verify routing tables
 - How traceroute reveals packet paths
 - How enterprise branch networks reduce routing complexity
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Challenge Extensions (Optional)

Challenge 1: Replace Default Routes with Static Routes

Remove:

```
no ip route 0.0.0.0 0.0.0.0 10.1.12.1
```

Configure explicit routes instead.

Compare routing tables.

Challenge 2: Add an ISP Router

Add a fourth router:

R2

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R1(HQ) ---- ISP

/

/
R3

Configure:

```
ip route 0.0.0.0 0.0.0.0 ISP-IP
```

on HQ.

Observe how branch traffic automatically reaches the ISP through HQ.

Challenge 3: Create a Floating Static Route

Configure a backup route:

```
ip route 192.168.30.0 255.255.255.0 10.1.12.2 10
```

Then shut down the primary link and observe failover behavior.

Challenge 4: Analyze the Routing Table

Identify:

- Connected Routes (C)
- Static Routes (S)
- Default Routes (S*)
- Local Routes (L)

Explain why each route exists.